

---

# CROSS-MODAL ZERO-SHOT LEARNING FOR DECODING OF VISUAL NEURAL REPRESENTATIONS

---

COMP2261 MACHINE LEARNING COURSEWORK

**Ahmed Latif**

Durham University  
Durham, DH1 3LE, UK  
ahmed.latif@durham.ac.uk

December 23, 2025

## ABSTRACT

Traditionally, classification on EEG signal data has occurred under supervised paradigms, where unseen trials are extensively labeled. The drawback of this however is that collecting lots of labeled EEG data is expensive, and impractical for the task of learning EEG stimuli response for every visual stimuli. In this report, we use a cross-modal zero-shot learning (ZSL) setup that predicts unseen classes by learning linear mappings from EEG features to pre-computed text and image embedding spaces. Based on initial data analysis showing unstable covariance (ill-conditioning) and a stronger relationship between EEG and visual embeddings than text, we choose to apply Ridge Regression and tune simple multimodal fusion to reduce noise and improve generalisation. On the ThingsEEG-Text dataset, the final model reaches an aggregated Top-1 accuracy of 10.65% ( $\times 21$  chance level), and consistently beats an Ordinary Least Squares baseline across subjects. We conduct an error analysis showing mistakes are not random: the model mainly confuses classes whose prototypes are close in the embedding space. This report aims to highlight that regularised linear mappings can still be effective for decoding unseen visual stimuli.

## 1 Introduction

Traditionally, using machine learning to classify unseen data has required extensive labeling under a supervised learning paradigm. This is effective for specific tasks, but limits visual stimuli based EEG decoding where collecting labeled data for every visual class is impractical. Advanced paradigms such as Zero-Shot Learning (ZSL) however can classify unlabeled, unseen data, by instead classifying a corresponding semantic which mapped from an EEG trial.

To gain an understanding over the field, we opt to review current literature detailing recent developments. Recent research has aimed to use Deep-Learning based neural networks to learn EEG signal patterns for classification. We see in [Du et al., 2023] visual neural representations, their approach is to map EEG signals to multimodal features using trained neural-networks to classify, processing the semantic data using CLIP and CORnet-S networks to give embeddings. This approach, while yielding significantly-above chance results, shows that even in research, achieving Top-1 classification results above 15% is difficult, even when utilizing deep learning models.

For this report, I introduce a cross-modal Zero-Shot learning paradigm for the task of decoding unseen, unlabeled EEG signals by linearly mapping modalities to each other. I aim to show that through utilizing per-dataset modality characteristics and cross-modal correlations, we can apply zero-shot learning to this problem.

## 2 Data and Paradigm

### 2.1 Dataset overview

We use the ThingsEEG-Text dataset, which provides EEG recordings alongside visual and textual embeddings, with a predefined split between seen and unseen classes. EEG uses a fixed post-stimulus window (70-400 ms), being downsampled (1000->100 Hz), and uses 17 channels, while the visual set is provided as CORnet-S embeddings and the textual set as CLIP text embeddings (Du et al. [2023]). Key dataset statistics can be seen in Table 1.

Table 1: Per-subject dataset statistics for ThingsEEG-Text dataset across all modalities seen and unseen splits.

| Split  | Classes | Trials | Trials per class | Ch. | Timepoints | Text dim | Visual dim |
|--------|---------|--------|------------------|-----|------------|----------|------------|
| Seen   | 1654    | 16540  | 10               | 17  | 100        | 512      | 1000       |
| Unseen | 200     | 16000  | 80               | 17  | 100        | 512      | 1000       |

### 2.2 Problem definition

For a single subject, let the *seen* split be defined by  $D_{\text{seen}} = \{(b, v, t, y) \mid b \in B_{\text{seen}}, v \in V_{\text{seen}}, t \in T_{\text{seen}}, y \in Y_{\text{seen}}\}$  denote the collection of *seen*-class samples, where  $B_{\text{seen}}$  are brain features,  $V_{\text{seen}}$  are visual features,  $T_{\text{seen}}$  are textual features, and  $Y_{\text{seen}}$  is the set of seen labels. Brain features for unseen classes,  $B_{\text{unseen}}$ , are not used during training and are only accessed at inference for classification and accuracy scoring. The label sets are *mutually exclusive* such that  $Y_{\text{seen}} \cap Y_{\text{unseen}} = \emptyset$ .

As all EEG trials are time-aligned, we flatten each  $(17 \times 100)$  channel-time matrix into a single 1700-D feature vector, giving

$$\mathbf{B}_{\text{seen}} \in \mathbb{R}^{16540 \times 1700}, \quad \mathbf{B}_{\text{unseen}} \in \mathbb{R}^{16000 \times 1700}.$$

The corresponding visual and textual embeddings are

$$\mathbf{T}_{\text{seen}} \in \mathbb{R}^{16540 \times 512}, \quad \mathbf{T}_{\text{unseen}} \in \mathbb{R}^{16000 \times 512}, \quad \mathbf{V}_{\text{seen}} \in \mathbb{R}^{16540 \times 1000}, \quad \mathbf{V}_{\text{unseen}} \in \mathbb{R}^{16000 \times 1000}.$$

### 2.3 Data exploration

To prevent any leakage, all data exploration is conducted exclusively on  $D_{\text{seen}}$ .

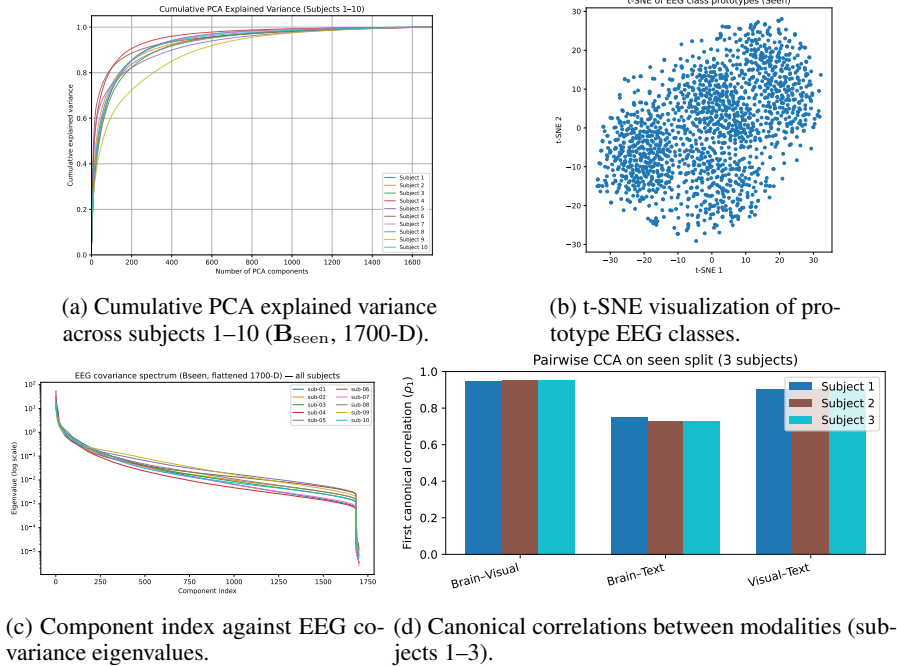


Figure 1: Exploratory data analysis of EEG representations: variance structure, centering behavior, clustering, and cross-modal correlation.

In **Figure (a)**, there is no clear low-dimensional elbow. Retaining 95% of the variance requires  $\sim 500$  principal components, suggesting variance is broadly distributed, and PCA provides limited dimensionality reduction.

**Figure (b)** shows the neighborhood structure of centered EEG class prototypes via t-SNE for 1 subject. The lack of clustering is suggestive of limited class separability in EEG feature space.

**Figure (c)** shows the EEG covariance spectrum decaying slowly, and spanning several orders of magnitude, indicating variance is spread across many directions rather than concentrated in a small subspace (supported by figure (a)). The tail of near-zero eigenvalues suggests ill-conditioned, highly correlated features.

**Figure (d)** shows the first canonical correlation obtained by applying CCA between each modality in  $D_{\text{seen}}$  (sampled to 3 subjects due to computational restraints). We observe strong linear alignment, particularly between the  $(\mathbf{B}_{\text{seen}}, \mathbf{V}_{\text{seen}})$  pairing in comparison to  $(\mathbf{B}_{\text{seen}}, \mathbf{T}_{\text{seen}})$ , suggesting that visual embeddings share a strong linear relationship with EEG responses.

## 2.4 Zero-Shot proposal

For this particular dataset, given the strong correlation between the modalities and the lack of distinct class clusters in  $\mathbf{B}_{\text{seen}}$  (see Figure 1), we opt to use a cross-modal, embedding-based zero-shot learning paradigm.

Zero-shot learning also has clear practical value in real-world EEG recording settings. Collecting labeled EEG data for every new category is expensive and time-consuming. By mapping EEG signals to visual and textual embedding spaces, unseen categories can be recognized via their class representations without additional EEG data collection.

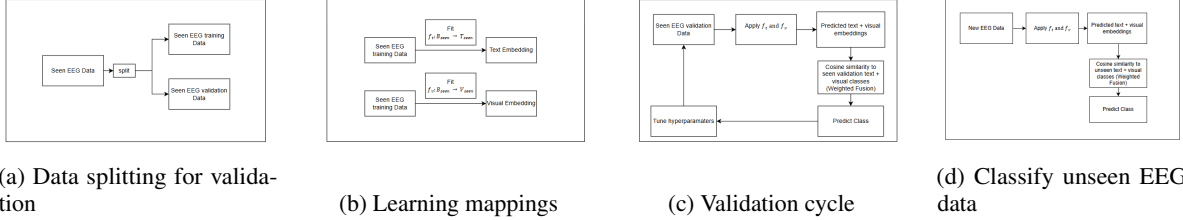


Figure 2: Zero-shot learning paradigm.

### 2.4.1 Training/Validation/Testing

$B_{\text{seen}}$  is split into training and validation subsets, splitting by class ID so we can fix hyperparameters. Hyperparameters are selected based on classification results from the training split against the validation split, choosing fixed hyperparameters which maximise accuracy for inference. Inference is evaluated once, where at test time,  $B_{\text{unseen}}$  is mapped to a predicted visual and textual embedding. We then classify such that:

$$\hat{y} = \underset{c \in \{1, \dots, 200\}}{\operatorname{argmax}} \left( w_t \cos(\hat{\mathbf{t}}, \mathbf{p}_c^t) + w_v \cos(\hat{\mathbf{v}}, \mathbf{p}_c^v) \right), \quad w_t + w_v = 1$$

where  $p_c$  is the class mean prototype embedding of the unseen class  $c$  for textual or visual embeddings.

## 2.5 Limitations

The seen split only has 10 trials per class, which may lead to noisy class estimates. Additionally, the strong linear alignment seen between EEG and visual embeddings may be specific to the **brain\_feature** and **visual\_feature** datasets. Additionally, as the dataset is already preprocessed at the feature level, We choose to avoid extra preprocessing that could alter the cosine-similarity structure in prototype ZSL (see Sec 2.1). This decision could mean that per-subject bias may impact the final classification results.

## 3 Model Development

### 3.1 Baseline task definition

Utilizing the linear relationships seen in Fig 1, we train two separate linear ordinary least squares regressors using `sklearn.linear_model.LinearRegression`. Each regressor learns the mappings

$$f_t : \mathbf{B}_{\text{seen}} \rightarrow \hat{\mathbf{T}}_{\text{seen}}, \quad f_v : \mathbf{B}_{\text{seen}} \rightarrow \hat{\mathbf{V}}_{\text{seen}}.$$

where we also fit for the intercept to counter subject specific bias (Fig 1). Class prototypes are then constructed for the unseen visual and textual embeddings, such that a single averaged embedding represents each class where

$$I_c = \{i \mid y_i = c\}, \quad p_c^t = \frac{1}{|I_c|} \sum_{i \in I_c} t_i, \quad p_c^v = \frac{1}{|I_c|} \sum_{i \in I_c} v_i$$

Collecting  $\{p_c\}_{c=1}^{200}$  row-wise gives  $\mathbf{P}_t \in \mathbb{R}^{200 \times 512}$  and  $\mathbf{P}_v \in \mathbb{R}^{200 \times 1000}$  with  $|I_c| = 80$  (see Table. 1).

Given  $b \in B_{\text{unseen}}$ , we can predict  $\hat{t} = f_t(b) \in \mathbb{R}^{512}$  and  $\hat{v} = f_v(b) \in \mathbb{R}^{1000}$ . We then perform similarity scoring through zero-shot classification, where the cosine similarity scores for visual and textual embeddings are computed as:

$$\mathbf{S}^{(t)} = \cos(\mathbf{P}_t, \hat{\mathbf{t}}) \in \mathbb{R}^{200} \quad \mathbf{S}^{(v)} = \cos(\mathbf{P}_v, \hat{\mathbf{v}}) \in \mathbb{R}^{200}$$

producing a vector of cosine similarity scores between each prototype in the prototype matrix and the prediction.

We combine modality cosine scores such that  $\mathbf{S} = \mathbf{S}^{(t)} + \mathbf{S}^{(v)}$  and predict the unseen EEG trial using the fused cosine scoring rule defined 2.4.1, setting equal weights for fusion.

As a linear approach trained on seen data with no hyperparameters set, the baseline can be unstable against ill-conditioned data (See Fig 1) and does not take into account each modalities contribution to the accuracy.

### 3.2 Improved model

#### 3.2.1 Hypothesis

As  $\mathbf{B}_{\text{seen}} \in \mathbb{R}^{16540 \times 1700}$  has variance split across many dimensions and no clear low-dimensional elbow, and an eigenvalue spectrum with a tail of near-zero modes (Fig 1), we suggest replacing OLS with ridge will stabilize the EEG -> Visual+Textual mappings:  $f_t, f_v$ , by suppressing ill-conditioned eigenvalues in the feature space (Hoerl and Kennard [1970]).

Additionally, our CCA shows stronger linear relationships for  $(B_{\text{seen}}, V_{\text{seen}})$  compared to  $(B_{\text{seen}}, T_{\text{seen}})$  (Fig 1), we hypothesize a visual-heavy fusion ( $w_v > w_t$ ) in the cosine prototype rule of 2.4.1 will reduce near-neighbour confusions by mainly classifying on the modality whose embedding is most linearly aligned with the EEG data.

#### 3.2.2 Train/Validation split

To tune ridge's  $\alpha$  without accessing  $\mathbf{B}_{\text{unseen}}$ , we split  $D_{\text{seen}}$  using a **90/10 class-wise split**, holding out enough validation data to classify while preserving training data for fitting the ridge maps. We form  $(\mathbf{B}_{\text{train}}, \mathbf{T}_{\text{train}}, \mathbf{V}_{\text{train}})$  which contain 90% of  $D_{\text{seen}}$ , and  $(\mathbf{B}_{\text{val}}, \mathbf{T}_{\text{val}}, \mathbf{V}_{\text{val}})$  with the remaining 10% where labels  $Y_{\text{seen}} = Y_{\text{train}} \cup Y_{\text{val}}$ ,  $Y_{\text{train}} \cap Y_{\text{val}} = \emptyset$ .

#### 3.2.3 Ridge Regression Formalization and $\alpha$ Selection

Let  $\mathbf{X} = \mathbf{B}_{\text{train}} \in \mathbb{R}^{n \times p}$  and  $\mathbf{Y} \in \{\mathbf{T}_{\text{train}}, \mathbf{V}_{\text{train}}\} \in \mathbb{R}^{n \times d}$ . Ridge regression solves:

$$\min_{(\mathbf{W}, \mathbf{b})} \|\mathbf{X}\mathbf{W} + \mathbf{1}\mathbf{b}^T - \mathbf{Y}\|_F^2 + \alpha \|\mathbf{W}\|_F^2 \quad \alpha > 0,$$

where  $\mathbf{1} \in \mathbb{R}^n$  is the all-ones vector and  $\|\cdot\|_F$  is the Frobenius norm.

To center for bias, we define column means and centered matrices

$$\mu_{\mathbf{X}} = \frac{1}{n} \mathbf{1}^T \mathbf{X} \in \mathbb{R}^p, \quad \mu_{\mathbf{Y}} = \frac{1}{n} \mathbf{1}^T \mathbf{Y} \in \mathbb{R}^d \quad \mathbf{X}_c = \mathbf{X} - \mathbf{1}\mu_{\mathbf{X}}, \quad \mathbf{Y}_c = \mathbf{Y} - \mathbf{1}\mu_{\mathbf{Y}}$$

Then the best ridge satisfies the linear system

$$(\mathbf{X}_c^T \mathbf{X}_c + \alpha \mathbf{I}_p) \mathbf{W} = \mathbf{X}_c^T \mathbf{Y}_c$$

Giving us

$$\mathbf{W} = (\mathbf{X}_c^T \mathbf{X}_c + \alpha \mathbf{I}_p)^{-1} \mathbf{X}_c^T \mathbf{Y}_c, \quad b = \mu_{\mathbf{Y}} - \mu_{\mathbf{X}} \mathbf{W}$$

We then tune  $\alpha$  per subject, punishing large weights. For each candidate  $\alpha$ , we use a log grid  $A = \{1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, 200, 300, 500, 1000\}$ , as this is consistent with  $\alpha$  selection practice ([Friedman et al., 2010]). We fit two ridge models:

$$f_t^{(\alpha)} : \mathbf{B}_{\text{train}} \rightarrow \hat{\mathbf{T}}_{\text{train}} \quad f_v^{(\alpha)} : \mathbf{B}_{\text{train}} \rightarrow \hat{\mathbf{V}}_{\text{train}}$$

we then compute prototypes on  $(\mathbf{T}_{\text{val}}, \mathbf{V}_{\text{val}})$ , produce a cosine similarity score vector  $\mathbf{S}_{\alpha} = \mathbf{S}_{\alpha}^{(t)} + \mathbf{S}_{\alpha}^{(v)}$  on validation class prototypes and validation predictions (see 3.1 for reference) and select  $\alpha$  such that

$$\alpha^* = \underset{\alpha \in A}{\operatorname{argmax}} \operatorname{Acc}_{\text{ZSL}}(\mathbf{S}_{\alpha})$$

Where we set a singular  $\alpha^*$  for both modalities, as ill-conditioning seen in Fig 1 lives in  $\mathbf{B}_{\text{seen}}$ . We use a set of candidate ratios  $\mathcal{R} = \{(w_t, w_v) \mid w_t \in \{0, 0.1, \dots, 1\}, w_v = 1 - w_t\}$  when classifying  $(\hat{\mathbf{T}}_{\text{val}}, \hat{\mathbf{V}}_{\text{val}})$ . Across a sample of 3 subjects, we find that the highest consistently scoring combination of weights for the train/val split is  $(w_t, w_v) = (0.1, 0.9)$ , which is consistent with the EEG-Visual linearity score we observed in the Fig 1 CCA chart. From here on out, we set these as our fixed, permanent weights for ridge globally.

#### 3.2.4 Fusion and Inference

Once the optimal alpha  $\alpha^*$  has been selected, we retrain on the full seen split such that

$$f_t^{(\alpha^*)} : \mathbf{B}_{\text{seen}} \rightarrow \hat{\mathbf{T}}_{\text{seen}} \quad f_v^{(\alpha^*)} : \mathbf{B}_{\text{seen}} \rightarrow \hat{\mathbf{V}}_{\text{seen}}$$

visual and textual predictions are then scored against prototypes via cosine similarity, and then classified using the inference rule stated in 2.4.1 such that

$$\mathbf{S} = 0.1 \cos(\mathbf{P}_t, \hat{\mathbf{t}}) + 0.9 \cos(\mathbf{P}_v, \hat{\mathbf{v}}), \quad \hat{c} = \underset{c \in \{1, \dots, 200\}}{\operatorname{argmax}} (\mathbf{S}_c).$$

## 4 Evaluation

### 4.1 Overall zero-shot performance

We evaluate zero-shot classification on the unseen split across all 10 subjects, giving 160,000 classified trials total (16,000 per subject). With Top-200 classification, chance Top-1 is 0.5%. The improved model achieves an aggregated Top-1 accuracy of 10.65%, a  $\sim 21\times$  chance-level performance. This supports the paradigm assumption that unseen-class recognition can be achieved by linearly mapping EEG trials into a multimodal semantic space (Section 2.4.1), selecting nearest class prototypes under cosine similarity with fusion. Table 2 shows that improvement is consistent across all

subjects, and is seen in Top-5 performance too; it also shows the greatest improvement being the weighted fusion, supporting our hypothesis in Sec 3.2.1.

Table 2: 3-model ablation across 10 subjects. Top-1 / Top-5 accuracy (%).

| Model           | Sub-01 |       | Sub-02 |       | Sub-03 |       | Sub-04 |       | Sub-05 |       | Sub-06 |       | Sub-07 |       | Sub-08 |       | Sub-09 |       | Sub-10 |       |
|-----------------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|
|                 | Top-1  | Top-5 | Top-1  | Top-5 | Top-1  | Top-5 | Top-1  | Top-5 | Top-1  | Top-5 | Top-1  | Top-5 | Top-1  | Top-5 | Top-1  | Top-5 | Top-1  | Top-5 | Top-1  | Top-5 |
| A (LinearReg)   | 7.41   | 21.89 | 6.07   | 19.05 | 7.28   | 21.80 | 5.54   | 18.40 | 5.55   | 17.38 | 7.72   | 22.63 | 8.51   | 23.64 | 9.24   | 26.39 | 5.76   | 18.15 | 8.61   | 24.52 |
| B (+Ridge)      | 7.68   | 22.45 | 7.83   | 22.71 | 8.17   | 23.59 | 7.64   | 22.11 | 6.09   | 18.53 | 9.44   | 26.54 | 9.97   | 27.74 | 11.06  | 29.66 | 6.74   | 19.97 | 11.26  | 29.92 |
| C (+1/9 fusion) | 11.48  | 30.02 | 8.89   | 25.01 | 9.29   | 26.20 | 8.59   | 24.65 | 7.71   | 22.75 | 11.96  | 31.71 | 12.46  | 32.20 | 14.01  | 35.20 | 8.01   | 22.66 | 14.07  | 32.99 |

| Model           | P <sub>macro</sub> (%) | R <sub>macro</sub> (%) | F1 <sub>macro</sub> (%) |
|-----------------|------------------------|------------------------|-------------------------|
| A (LinearReg)   | 10.49 ± 2.84           | 7.17 ± 1.30            | 5.52 ± 1.19             |
| B (+Ridge)      | 12.86 ± 3.78           | 8.59 ± 1.67            | 6.44 ± 1.48             |
| C (+1/9 fusion) | 12.33 ± 3.51           | 10.65 ± 2.31           | 8.94 ± 2.23             |

Table 3: Macro metrics across 10 subjects (mean ± std, %).

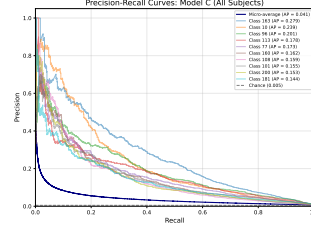


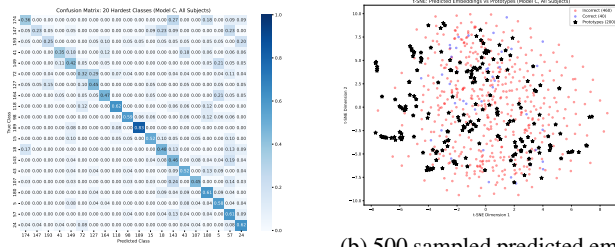
Figure 3: Precision-recall curves for Model C (all subjects).

Ridge improves balance performance, addressing the ill-balanced eigenvalues in Fig 1, and fusion yields higher recall confirming our hypothesis in 3.2.1(best macro F1 score). This suggests gains came from retrieving more true positives rather than being more conservative. Figure 3 reflects this: precision-recall is above chance but still low overall, but a subset of classes achieves substantially higher AP, indicating the model is really good at classifying a subset of the total classes.

## 4.2 Error Analysis

Sampling some Top-5 inference outcomes from the 160,000 total unseen predictions, we see that correct predictions tend to occur under close prototype competition: (e.g., 173->173 Top-5 [173,154,136,69,35], margin 0.0260), where margins are small (0.016-0.06), indicating the correct prediction is only narrowly ahead of close alternatives. Misclassifications reflect this, where a competing prototype marginally outranks the true class (e.g., 108->75 Top-5 [75, 79, 108, 53, 160] margin 0.0628), although true values of misclassified EEG trials occasionally aren't present within the Top-5 list, suggesting occasional misplacement rather than a pure near-tie, which can be attributed to the small sample size per class (Sec 2.5).

Taking a wider look at our model, the t-SNE plot in 4b show the predicted embeddings forming a cloud, where the predictions are primarily made in regions with prototypes in crowded classes. Taking a look at 4a, we see errors aren't distributed uniformly, but rather concentrated in specific cells, indicating incorrect predictions repeatedly identify a similar alternative prototype. Table 4 confirms this; the most frequent mistakes recur as the same true->predicted pairs, and the pairs tend to have a high cosine similarity score, with a positive Pearson correlation  $\approx 0.35$  between prototype similarity and confusion frequency. Together, these figures indicate the model's tenancy to incorrectly classify due to near-neighbor prototype overlap, and outlines ridge regressions limitation of struggling with rare/outlier classes.



(a) 20 hardest unseen classes

(b) 500 sampled predicted embeddings vs prototypes

|            | P1     | P2     | P3     | P4     | P5     | P6     | P7     | P8     | P9     | P10    |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| True       | 113.00 | 20.00  | 27.00  | 123.00 | 12.00  | 143.00 | 55.00  | 133.00 | 116.00 | 80.00  |
| Prediction | 163.00 | 10.00  | 163.00 | 163.00 | 39.00  | 163.00 | 163.00 | 39.00  | 181.00 | 163.00 |
| Count      | 291.00 | 247.00 | 214.00 | 210.00 | 181.00 | 175.00 | 167.00 | 162.00 | 158.00 | 156.00 |
| Cos sim.   | 0.47   | 0.48   | 0.34   | 0.32   | 0.60   | 0.37   | 0.48   | 0.36   | 0.46   | 0.29   |

Table 4: Top-10 confusion pairs aggregated over 10 subjects (160,000 unseen trials), Pearson  $r=0.3497$ .

## References

- Changde Du, Kaicheng Fu, Jinpeng Li, and Huiguang He. Decoding visual neural representations by multimodal learning of brain-visual-linguistic features. *IEEE Trans. Pattern Anal. Mach. Intell.*, 45(9):10760–10777, 2023.
- A. Hoerl and R. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1): 55–67, 1970.
- Jerome Friedman, Trevor Hastie, and Robert Tibshirani. Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1):1–22, 2010. doi:10.18637/jss.v033.i01.